



ÉCOLE NATIONALE DES
PONTS
ET CHAUSSÉES



IP PARIS

Advanced Master in Railway and Urban Transport System Engineering - 2024

Ecole Nationale des Ponts et Chaussées

CREATION OF A SUBURBAN SERVICE IN KHARTOUM, SUDAN

Pre-opportunity study for « Khartoum Express Railways »



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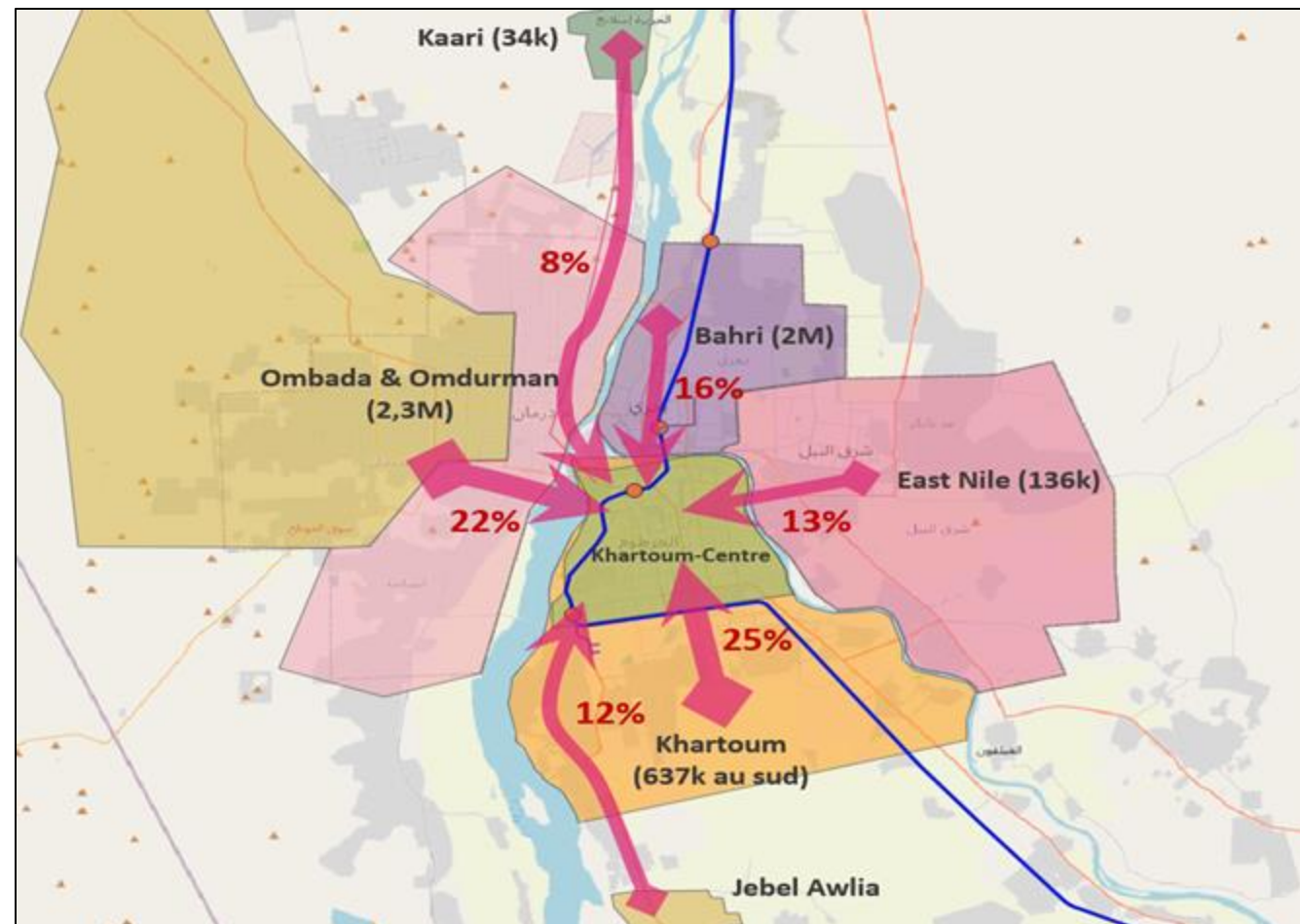
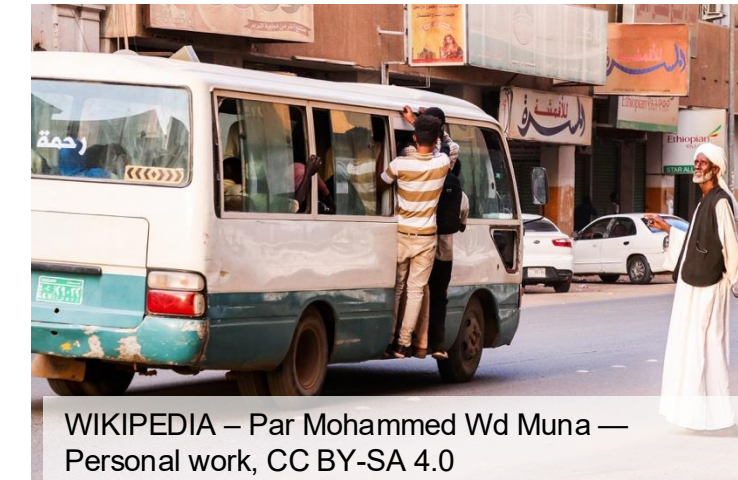
ABSTRACT

“This study focuses on the creation of a suburban service in Khartoum, capital of Sudan. The concentration of points of interest in the city center and the decentralization of the growing population in the suburbs make a new transportation solution necessary.

The study include a diagnosis of the situation, followed by a multi-criteria analysis to choose a solution. This solution is then sized, and its implementation is detailed in terms of infrastructure, rolling stock, and operation. Finally, a socio-economic assessment of the solution is carried out, including a detailed risk analysis.

The result is a complete rehabilitation of the existing railway line. The new service will offer frequent trains with new hybrid rolling stock. All equipment will be designed to allow a gradual increase in passenger flows, thus anticipating the population's growing adoption of the new rail solution.”

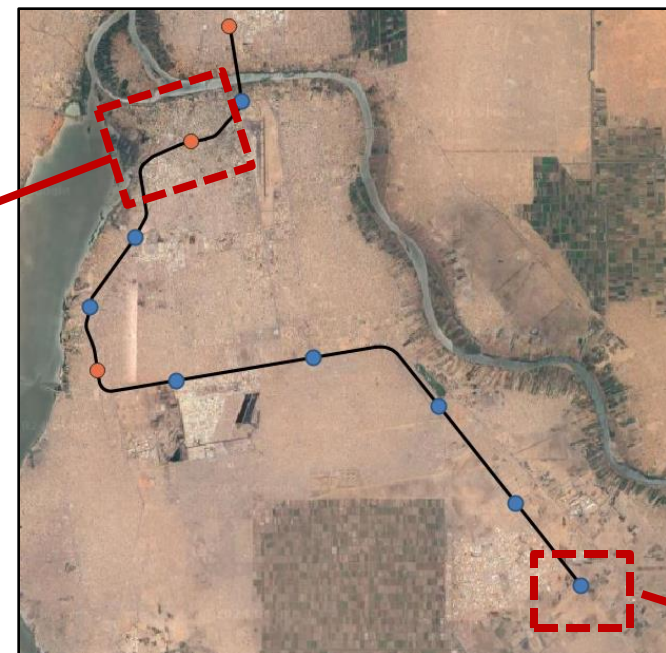
- **Sudan** : 47,9 M inhabitants, GDP/Capita: 997€
- **Khartoum** : 7M inhabitants, significantly increasing
- **Mean income** : ~60€/month of which 30% allocated to the public transport budget
- **Motorisation rate** : 4% (very low)



Mobility in Khartoum:

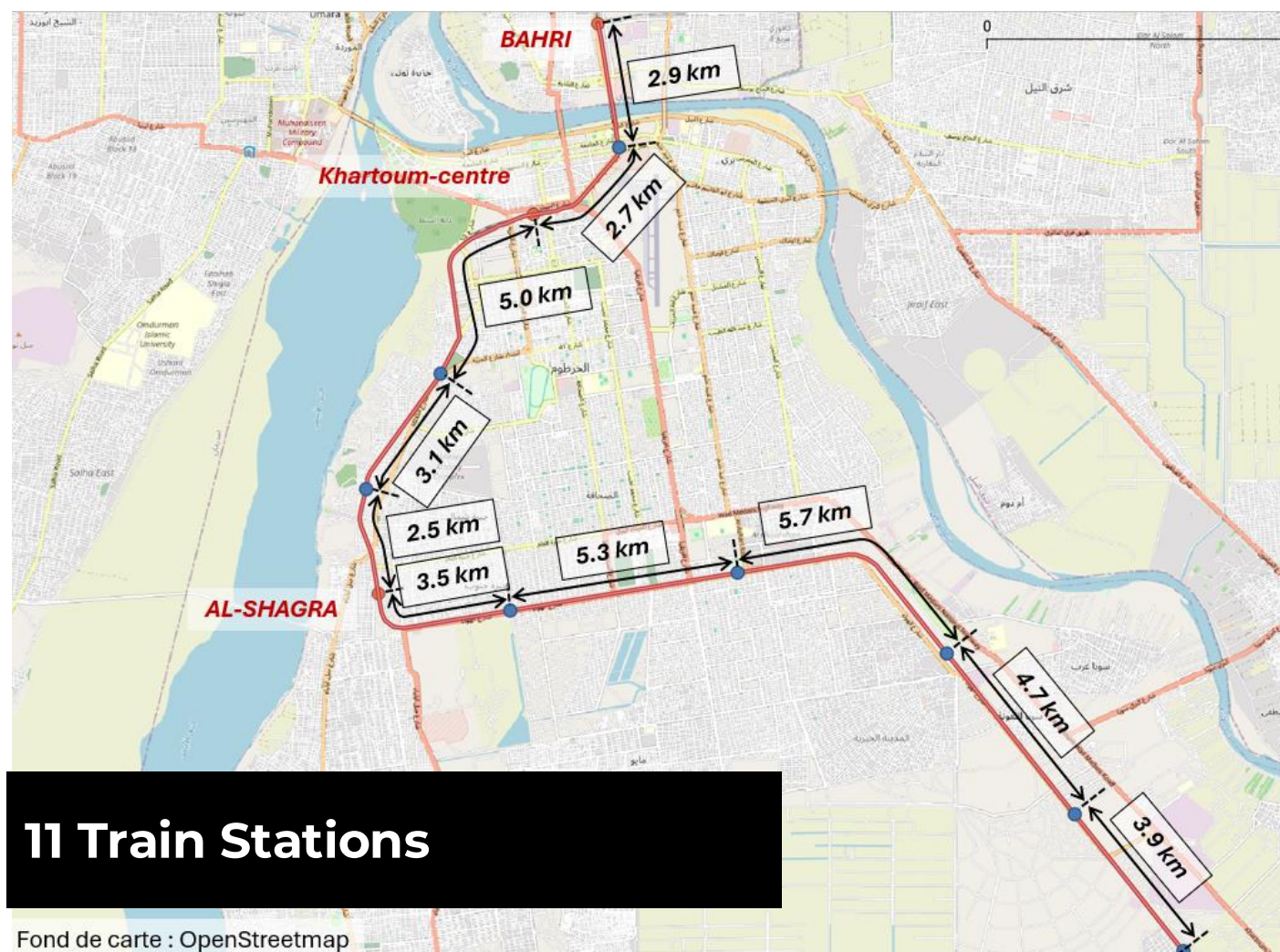
- Heavy congestion in the center and to cross the Nile
- Large use of minibuses causing considerable discomfort
- High level of passenger's bulk breaking
- Long daily journeys for essential travels

The study provides a modern mobility solution offering more capacity, safety and availability.



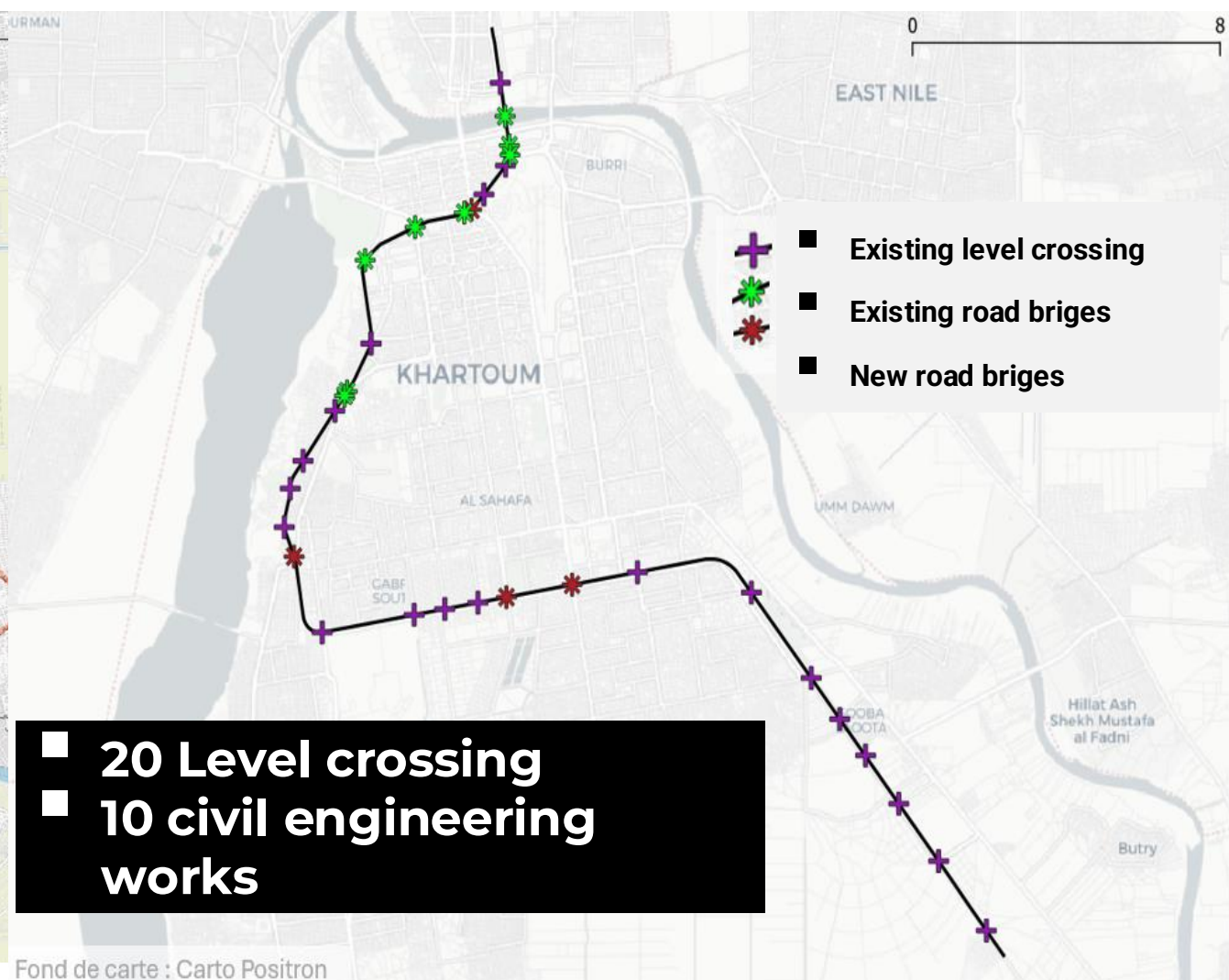
Rehabilitation of existing Maintenance workshop

1 New Maintenance workshop



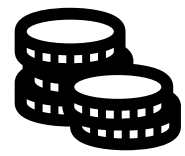
11 Train Stations

Fond de carte : OpenStreetmap



- Existing level crossing
- Existing road bridges
- New road bridges

Fond de carte : Carto Positron



	Total Cost (EUR)	Cost per km (EUR)
CAPEX	1,6 bn€	40 M€ / km
OPEX	89 M€/year	2,2 M€ / km / year
NPV	Negative balance compensated by socio-economic externalities.	

	Cost	% of global Cost
AFD funds	200 M€	12 %
World Bank funds	1,4 bn€	88 %



39 Km in 57 min (South to Khartoum Center)



11 Stations (5 multimodal stations)



4 Trains per hour, 40 km/h (max 80km/h)
Double track



127 500 Passenger per day
PPHPD: 3750



Hybrid Regiolis Train (Coradia)

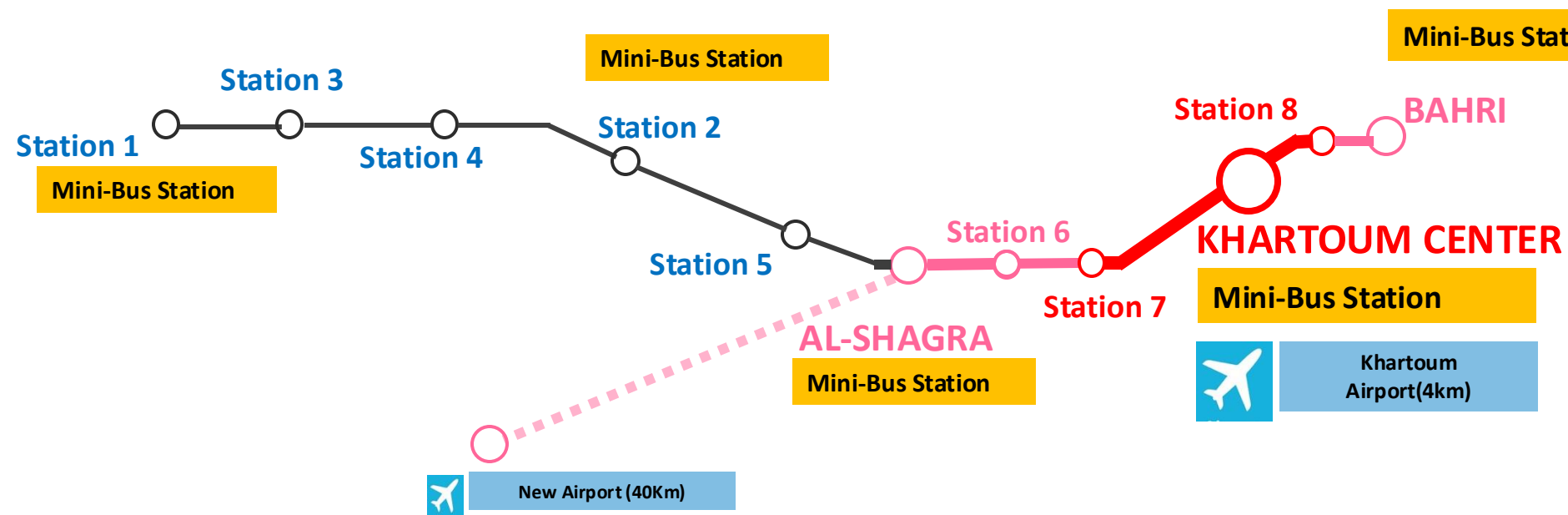


40 M€/Km



Project duration: 5 years

« Khartoum Express Railways » Line Plan :



*All wishes for peace among
Sudan and rebuilding the longest
railway network in Africa again !*